

ECAF submission content David check

Workflow

- German credit dataset (1000 instances, 20 features) preprocessed
- Made predictions on test set of 300 instances by training a random forest model without inclusion of protected attributes sex, age and foreign_worker. AUC = 77.78, accuracy = 74.5
- Resulted in 34 adversely predicted instances (“bad” class)
- For these 34 instances, SHAP values were generated with treeSHAP algorithm
- 6 providers (grok, openai, deepseek, gemini, claude and mistral) were asked to generate an explanatory narrative for the 34 adversely predicted instances (see example prompt in appendix)
- 3 providers (grok, deepseek, openai) were asked to extract certain information from the narratives. A majority voting function was applied to results in a final “extraction” JSON file for each instance for each generator LLM.
- This majority voted extraction was compared with a ground truth file
- From this comparison and absolute counts result the tables.

Repo (very messy): <https://github.com/WannesVandenBulck/ECAFairness>

Tables in mail attachment

- Table 1: raw metrics for each provider without protected attributes split.
- Table 2: metrics for each provider split by sex protected attribute
- Table 3: metrics for sex protected attribute averaged over all providers

Appendix

A. Generator prompt for adversely predicted instance 5

A machine learning model predicted that a loan applicant represents a BAD CREDIT RISK and therefore their loan application was DENIED.

YOUR TASK: Translate the following technical information into a clear, non-technical narrative explanation that helps the applicant understand:

- Why the model rejected their application in specific terms of their features
- Which factors were most important in this decision
- How their specific situation compared to typical applicants who were approved

INFORMATION YOU WILL RECEIVE:

1. DATASET INFORMATION: Context about the dataset, target variable and ML task used to train the model
2. TECHNICAL EXPLANATION METHOD: How we measure feature importance (SHAP values)
3. APPLICANT PROFILE: The applicant's specific feature values with comparisons to approved applicants averages and distributions
4. FEATURE IMPORTANCE ANALYSIS: SHAP values showing which features most influenced the decision
5. CLEAR INSTRUCTIONS: What narrative you should write

1. DATASET INFORMATION

The dataset contains information from the 1970s in Germany on a series of debtors that took a loan from the bank. It includes detailed categorical and numerical variables about their financial situation. Protected attributes age and sex were not used to make the machine prediction. Keep in mind that at the time Germany used Deutsche Marks (DM) with an average yearly salary of 10,000 to 20,000 DM

Target Variable: The target variable indicates whether the customer is a bad credit risk (1) or a good credit (0). We are predicting the bad credit class.

ML Task: The ML model aims to predict whether a new customer will be a bad credit risk

Protected attributes sex, age and foreign_worker were not used to make the machine prediction.

Target Encoding (target_credit):

- 1 (model predicted 1) = BAD CREDIT RISK: Applicant is predicted to be a bad credit risk (credit application was DENIED)
- 0 (not predicted, favorable) = GOOD CREDIT: Applicant showed good credit behavior (credit application was APPROVED)

2. TECHNICAL EXPLANATION: SHAP VALUES

You are given SHAP values for this applicant's prediction.

SHAP values explain how much each feature contributes to the model's prediction for this specific applicant.

Each feature has a SHAP value that tells you:

- How much that feature influenced the model's decision for this applicant.
- Whether it pushed the prediction toward "bad credit risk" (positive contribution) or "good credit" (negative contribution).
- Larger absolute values indicate features with stronger influence on the prediction.

Features are ranked by their absolute SHAP values, with the most influential features listed first.

Features with positive SHAP values contributed toward a "bad credit risk" prediction.

Features with negative SHAP values contributed toward a "good credit" prediction.

IMPORTANT: Only the SHAP values of the top 3 most important features are included in the table below. These are the features with the strongest influence on this applicant's prediction.

3. APPLICANT PROFILE

You are writing a narrative tailored to this specific person who is rejected a loan application.

Feature values (with comparisons to approved applicants where available):

- status = <0 DM - among approved applicants: none: 50.2%, 0-200 DM: 23.2%, <0 DM: 19.8%, ≥200 DM/salary: 6.8%
- Status of checking account
- duration = 28.00 (approved applicants avg: 19.11) - Duration of credit request in months
- credit_history = existing paid - among approved applicants: existing paid: 50.7%, critical/other credits: 34.9%, past delays: 8.9%, all paid at bank: 3.2%, no credits/all paid: 2.3%
- Credit history rating

- purpose = new car - among approved applicants: radio/TV: 32.3%, new car: 19.8%, furniture/equipment: 17.4%, used car: 11.8%, business: 9.3%, education: 4.6%, repairs: 1.9%, domestic appliances: 1.4%, retraining: 1.1%, others: 0.5% - Purpose of credit
- amount = 4006.00 (approved applicants avg: 2887.25) - Amount of credit requested in DM
- savings = <100 DM - among approved applicants: <100 DM: 55.6%, unknown/none: 19.5%, 100-500 DM: 10.4%, 500-1000 DM: 8.2%, ≥1000 DM: 6.3% - Savings account status
- employment_duration = 1-4 years - among approved applicants: 1-4 years: 34.9%, ≥7 years: 24.9%, 4-7 years: 19.6%, <1 year: 15.6%, unemployed: 4.9% - Employment duration
- installment_rate = 3.00 (approved applicants avg: 2.94) - Installment rate as percentage of disposable income
- sex = male - The sex of the applicant
- other_debtors = none - among approved applicants: none: 91.8%, guarantor: 5.4%, co-applicant: 2.8% - Other debtors/guarantors
- present_residence = 2.00 (approved applicants avg: 2.81) - Years at present residence
- property = car/other - among approved applicants: car/other: 33.9%, real estate: 30.9%, building society/life insurance: 23.0%, unknown/none: 12.3% - Property status
- age = 45.0 - The age of the applicant in years
- other_installment_plans = none - among approved applicants: none: 84.0%, bank: 11.8%, stores: 4.2% - Other installment plans
- housing = own - among approved applicants: own: 75.8%, rent: 15.3%, for free: 8.9% - Housing situation
- number_credits = 1.00 (approved applicants avg: 1.42) - Number of existing credits at this bank
- job = unskilled-resident - among approved applicants: skilled/official: 64.6%, unskilled-resident: 20.9%, management/self-employed/highly skilled: 12.8%, unemployed/unskilled-non-resident: 1.8% - Job type
- people_liable = 1.00 (approved applicants avg: 1.14) - Number of people financially dependent
- telephone = none - among approved applicants: none: 59.6%, yes registered: 40.4% - has a telephone or not
- foreign_worker = yes - Foreign worker or not

The model's prediction:

- Predicted probability of bad credit: 55.0%

4. FEATURE IMPORTANCE ANALYSIS (Ranked by Influence)

Feature SHAP_Value

status 0.126281

amount -0.108109

purpose 0.065853

5. YOUR NARRATIVE TASK

TASK:

Your goal is to generate a plausible textual explanation or narrative explaining why the loan application was denied for this applicant.

PERSONALIZATION INSTRUCTION:

Based on the provided PERSONAL INFORMATION about the applicant, create a personalized narrative tailored to them.

The narrative should feel like it was written specifically for this individual, acknowledging their personal circumstances and creating a more personalized experience.

However, do not force the personalization: it should be seamlessly integrated into the narrative.

Write a detailed narrative explanation tailored to this non-technical reader that MUST explain:

- 1) The current situation of the applicant (what are their features and background).
- 2) The model's predicted probability of bad credit and what this means for the applicant.
- 3) Why the application was denied, which features were most important in driving this prediction and why.
- 4) How each of the most important features contributed (either pushing toward bad credit or toward good credit).
- 5) What the applicant should do next

CONSTRAINTS:

- Do NOT invent new SHAP values or new numerical values.
- Do not use the numeric SHAP values in your answer. Instead, discuss the ranking and direction of influence.
- Do not talk about model internals, algorithms, or training details.
- Do not start with greeting or closing statements. Focus on the narrative.

STYLE:

- Length: 12-15 sentences.
- Write a coherent narrative without bullet points or tables. The goal is to have a plausible narrative/story.
- Directly address the applicant and provide PERSONALIZED insights tailored to THEIR situation (you can use the personal information provided), but let it sound natural.
- Do NOT copy-paste feature names, but instead incorporate them naturally in the narrative.
- Include feature values and their comparisons to averages or distributions, but reserve this for features where it really clarifies the explanation.

B. Extractor prompt for instance 5 for narrative generated by Claude

EXTRACTION TASK: Analyze the following narrative to extract prediction explanation data.

CONTEXT: An LLM generated this narrative to explain why a model predicted a negative outcome for this instance. You must extract information from the narrative to validate its accuracy.

1. DATASET & TARGET INFORMATION

Dataset: The dataset contains information from the 1970s in Germany on a series of debtors that took a loan from the bank. It includes detailed categorical and numerical variables about their financial situation. Protected attributes age and sex were not used to make the machine prediction. Keep in mind that at the time Germany used Deutsche Marks (DM) with an average yearly salary of 10,000 to 20,000 DM

Target: The target variable indicates whether the customer is a bad credit risk (1) or a good credit (0). We are predicting the bad credit class.

Task: The ML model aims to predict whether a new customer will be a bad credit risk

Features:

feature_name	feature_desc
status	Status of checking account
duration	Duration of credit request in months
credit_history	Credit history rating
purpose	Purpose of credit
amount	Amount of credit requested in DM
savings	Savings account status
employment_duration	Employment duration
installment_rate	Installment rate as percentage of disposable income
sex	The sex of the applicant
other_debtors	Other debtors/guarantors
present_residence	Years at present residence
property	Property status
age	The age of the applicant in years
other_installment_plans	Other installment plans
housing	Housing situation
number_credits	Number of existing credits at this bank
job	Job type
people_liable	Number of people financially dependent
telephone	has a telephone or not
foreign_worker	Foreign worker or not

Categorical Feature Mappings (translate mentions to codes):

ATTRIBUTE VALUE MAPPINGS

****status:****

- 1: <0 DM
- 2: 0-200 DM
- 3: ≥200 DM/salary

- 4: none

****credit_history:****

- 1: no credits/all paid
- 2: all paid at bank
- 3: existing paid
- 4: past delays
- 5: critical/other credits

****purpose:****

- 1: new car
- 2: used car
- 3: furniture/equipment
- 4: radio/TV
- 5: domestic appliances
- 6: repairs
- 7: education
- 8: retraining
- 9: business
- 10: others

****savings:****

- 1: <100 DM
- 2: 100-500 DM
- 3: 500-1000 DM
- 4: ≥1000 DM
- 5: unknown/none

****employment_duration:****

- 1: unemployed
- 2: <1 year
- 3: 1-4 years
- 4: 4-7 years
- 5: ≥7 years

****sex:****

- 0: male
- 1: female

****other_debtors:****

- 1: none
- 2: co-applicant
- 3: guarantor

****property:****

- 1: real estate
- 2: building society/life insurance
- 3: car/other
- 4: unknown/none

****other_installment_plans:****

- 1: bank
- 2: stores
- 3: none

****housing:****

- 1: rent

- 2: own
- 3: for free

****job:****

- 1: unemployed/unskilled-non-resident
- 2: unskilled-resident
- 3: skilled/official
- 4: management/self-employed/highly skilled

****telephone:****

- 1: none
- 2: yes registered

****foreign_worker:****

- 1: yes
- 2: no

2. WHAT TO EXTRACT

You must fill a JSON object with three sections:

SECTION 1: predicted_probability

Extract if explicitly mentioned in narrative (as decimal 0.0-1.0). Otherwise fill with "NaN".
Do not guess - only fill if clearly stated.

SECTION 2: most_important_features

These are the TOP 3 most important features the narrative emphasizes.

FOR EACH of the 3 ranks (1 to 3):

- name: The official feature name from the feature list above (exact match, case-sensitive).
Fill with: The actual feature name as stated in the feature list above

- sign: Whether this feature pushes toward NEGATIVE (1) or POSITIVE (-1) outcome.
Fill with: 1 or -1 only

- value:

The numeric value of a numeric feature for this instance as mentioned in narrative.

For categorical features, fill with the numeric code corresponding to the mentioned category (using mappings above).

If the narrative does not mention a value for this feature, fill with "NaN".

Fill with: A number, decimal, or "NaN" if not mentioned

- rank: Already filled (1, 2, 3, etc.) - DO NOT CHANGE

SECTION 3: features

ALL features from the narrative are listed here with their names ALREADY FILLED IN.

You ONLY need to fill:

- mentioned: 1 if this feature is discussed in narrative, 0 if not mentioned
Fill with: 0 or 1 only

- value: The value of a feature as mentioned in the narrative
Fill with: A number, decimal, code, or "NaN" if not mentioned

- name: ALREADY PRE-FILLED - DO NOT CHANGE

3. FORMAT RULES

- Numbers: No decimals for integers (24 not 24.0), decimals include point (0.87, 0.42)
- NaN: Use exact string "NaN" (not nan, null, or empty)
- Binary (0/1): Only 0 or 1, never true/false/yes/no
- Codes: Use numeric codes from mappings (not readable text)
- Feature names: MUST match exactly from feature list (case-sensitive)

4. TEMPLATE TO FILL (JSON FORMAT)

Fill all fields marked with * based on the narrative. Feature names in "features" are pre-filled.

```
```json
{
 "predicted_probability": "*",
 "most_important_features": [
 {
 "rank": 1,
 "name": "*",
 "sign": "*",
 "value": "*"
 },
 {
 "rank": 2,
 "name": "*",
 "sign": "*",
 "value": "*"
 },
 {
 "rank": 3,
 "name": "*",
 "sign": "*",
 "value": "*"
 }
],
 "features": [
 {
 "name": "status",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "purpose",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "duration",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "credit_history",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "amount",
 "mentioned": "*"
 }
]
}
```

```

 "value": "*"
 },
 {
 "name": "savings",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "employment_duration",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "installment_rate",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "other_debtors",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "present_residence",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "property",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "other_installment_plans",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "housing",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "number_credits",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "job",
 "mentioned": "*",
 "value": "*"
 },
 {
 "name": "people_liable",
 "mentioned": "*",
 "value": "*"
 },
}

```



```

{
 "name": "telephone",
 "mentioned": "*",
 "value": "*"
},
{
 "name": "age",
 "mentioned": "*",
 "value": "*"
},
{
 "name": "sex",
 "mentioned": "*",
 "value": "*"
},
{
 "name": "foreign_worker",
 "mentioned": "*",
 "value": "*"
}
]
}
...

```

## 5. NARRATIVE TO EXTRACT FROM

Your loan application was carefully reviewed, and the assessment found a 55% likelihood that you represent a credit risk — a figure that, while not dramatically high, was sufficient to result in a denial. This decision was shaped by a combination of factors specific to your financial profile, and understanding them may help you take steps toward a successful application in the future. The single most important factor working against your application was the current status of your checking account, which shows a negative balance. Among applicants who were approved, only about one in five had a negative checking account balance, while the majority either had no such account or maintained a positive balance. A negative checking account balance is a strong signal to lenders that day-to-day finances may already be under strain, making it harder to take on additional debt responsibly. The purpose of your loan — purchasing a new car — also contributed to the denial. While new car purchases are a common reason for borrowing, this category carries a somewhat higher perceived risk compared to, say, financing household electronics or furniture, and in combination with your other financial circumstances, it added to the concern. Interestingly, the amount you requested — 4,006 Deutsche Marks — actually worked in your favor to some degree. This figure is notably higher than the average loan amount among approved applicants, which sits around 2,887 DM, yet the model still registered this as a moderating factor rather than a purely negative one. That said, a larger loan amount in the context of financial stress can amplify risk, so it remains something lenders will scrutinize. Your loan duration of 28 months is also longer than the average of about 19 months among approved applicants, meaning you would be carrying this debt for a meaningfully extended period. Combined with your employment as an unskilled resident and relatively modest savings of under 100 DM, the overall picture suggests limited financial cushion if unexpected expenses arise. Your credit history of existing paid credits is a genuine positive, and your stable housing situation as a homeowner also reflects well on you. To improve your chances in a future application, the most impactful step would be to bring your checking account into a positive balance and maintain it there for several months before reapplying. Building up your savings, even modestly, would also strengthen your profile. You may also want to consider whether a shorter loan term or a slightly reduced loan amount could make your application more competitive.

## 6. FINAL INSTRUCTIONS

- The narrative might not state features explicitly but include descriptions that imply them.
- It is your task to extract the features the narrative includes by leveraging the feature descriptions and categorical feature mappings.
- Based on the feature names, categorical mappings, JSON format and these instructions, you must extract the relevant information from the narrative and fill in the JSON template accurately.

- If the narrative does not mention a specific piece of information, fill with "NaN" (for values) or 0 (for mentioned) as appropriate. Do not guess or infer beyond what is explicitly stated.
- Return ONLY the completed JSON in a code block, no other text or explanation, and do NOT change the template format